

ANALYSTLAB AFRICA

Data Science Internship Programme



Heart Disease Prediction Business Insights and Recommendation Report

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Project: Heart Disease Prediction
Dataset: Heart Disease Prediction Dataset

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Week 3 Submission – Advanced Data Analysis, Statistical Testing & Feature Engineering

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1. Introduction

This report presents the Business Insights and Recommendation Report for the Heart Disease Prediction project, prepared as part of the AnalystLab Africa Data Science Internship Programme, Week 3: Advanced Data Analysis, Statistical Validation & Feature Engineering.

The Week 3 brief requires a professional Business Insights Report summarising the project's findings for both technical and non-technical stakeholders, covering an executive summary, key analytical findings, business implications, evidence-based recommendations, dataset limitations, and next steps. This report brings together the evidence produced across the Week 3 workstream: the advanced exploratory data analysis, the Statistical Analysis Summary (six hypothesis tests and one correlation analysis), and the Feature Evaluation and Selection Summary, to translate that evidence into recommendations an organisation could act on.

The analysis is based on the Final_Modelling_Dataset, which builds on the cleaning and preprocessing completed in Week 2 and the feature engineering and evaluation completed earlier in Week 3. The dataset is now considered ready to be carried forward into Week 4 Machine Learning.

2. Executive Summary

This report details the exploratory data analysis, statistical validation, and feature engineering conducted on a heart disease dataset. The primary business problem is to develop a predictive model for heart disease, enabling early identification of at-risk individuals. The objective of this analysis was to understand the underlying patterns, key relationships between clinical features and heart disease, and to prepare a robust dataset for subsequent machine learning modelling.

Key findings indicate that Age, Maximum Heart Rate (MaxHR), Oldpeak (ST depression during exercise), ST-Slope, Exercise-Induced Angina, and Chest Pain Type are all statistically significant predictors of heart disease. Newly engineered features such as *Heart_Rate_Age_Ratio* and *ST_Exercise_Risk* further enhance predictive power by capturing nuanced interactions. The dataset is well-prepared, free from missing values and severe multicollinearity among numerical features, and is now ready for predictive modelling in Week 4.

3. Key Analytical Findings

3.1. Feature-Target Relationships

Note on units: MaxHR and Oldpeak means reported below are expressed in standardised (z-score) units, consistent with the StandardScaler transformation applied during Week 2 preprocessing, rather than in their original clinical units (beats per minute and millimetres of ST depression, respectively). A negative mean therefore indicates a value below the sample average, not a literal negative heart rate. This distinction should be made explicit if these figures are shared with a non-technical audience.

- **Age:** Statistically significant ($p < 0.001$) with heart disease. Patients with heart disease are, on average, older (mean 55.90 years) than those without (mean 50.55 years).
- **MaxHR:** Statistically significant ($p < 0.001$) with heart disease. Patients with heart disease exhibit lower standardised MaxHR values (mean -0.360) compared to healthy individuals (mean 0.446), indicating reduced cardiac capacity.
- **Oldpeak:** Statistically significant ($p < 0.001$) with heart disease. Higher standardised Oldpeak values (mean 0.369 in the heart disease group) are associated with heart disease, reflecting greater ST depression during exercise.
- **ST_Slope:** A strong association ($p < 0.001$) was found. Flat and Down-sloping ST segments are highly prevalent in heart disease patients (83% and 78% respectively), while Up-sloping is associated with lower prevalence (20%).

- **Exercise Angina:** Strongly associated ($p < 0.001$). 85% of patients experiencing exercise-induced angina have heart disease, compared to 35% without angina.
- **Chest Pain Type:** Significantly associated ($p < 0.001$). Asymptomatic chest pain has the highest heart disease prevalence (79%), highlighting the silent nature of the disease in some cases (Detrano et al., 1989).
- **FastingBS:** Significant association. 79% of patients with FastingBS > 120 mg/dL had heart disease, versus 48% in the ≤ 120 mg/dL group.
- **RestingBP & Cholesterol:** While clinically important, these features showed weaker individual correlations with heart disease in this dataset, but their distributions still differ between groups and they remain vital clinical indicators.

3.2. Engineered Feature Insights

- **Age_Group:** Categorisation of age (Young, Middle-Aged, Elderly) clearly showed increasing heart disease prevalence with age (Young: 33%, Middle-Aged: 54%, Elderly: 73%). This simplifies interpretation and captures non-linear effects.
- **BP_Category:** Categorisation of RestingBP (Normal, Elevated, Hypertension) revealed higher heart disease rates in the Hypertension group (63%) and Elevated group (50%).
- **Heart_Rate_Age_Ratio:** This engineered numerical feature (MaxHR / Age) showed a moderate negative correlation with heart disease ($r = -0.403$). A lower ratio consistently indicates a higher likelihood of heart disease, combining the predictive power of MaxHR and Age into a single, interpretable metric.
- **ST_Exercise_Risk:** This composite feature (ST_Slope + ExerciseAngina_Y) demonstrated very strong predictive power. Combinations such as 'Flat_Yes' (90.44% heart disease prevalence) and 'Down_Yes' (92.68%) are exceptionally high-risk, offering detailed risk stratification.

As noted in the accompanying Feature Evaluation and Selection Summary, ChestPainType, RestingECG and ST_Slope are human-readable reconstructions of variables already available in one-hot encoded form, rather than newly engineered information in the same sense as Age_Group, BP_Category, Heart_Rate_Age_Ratio and ST_Exercise_Risk (AnalystLab Africa, 2026b). This distinction does not change the business insights above but should be reflected in the final feature documentation.

3.3. Data Quality and Relationships

- **Missing Values:** No missing values were found, ensuring data completeness.
- **Multicollinearity:** Spearman correlation analysis revealed no strong multicollinearity ($r > 0.80$) among numerical features. This indicates that variables largely provide independent information, which is beneficial for model stability.
- **Normality:** Age, MaxHR, and Oldpeak were found to deviate from normality, justifying the use of non-parametric tests (Mann-Whitney U) for group comparisons (AnalystLab Africa, 2026c).
- **Data Leakage:** No data leakage was identified. All features are derived from information plausibly available at the time of diagnosis, ensuring the integrity of future predictive models (AnalystLab Africa, 2026b).

4. Business Implications

The analytical findings have several critical implications for healthcare organisations and decision-makers:

1. **Enhanced Risk Identification:** The identified key predictors (Age, MaxHR, Oldpeak, ST-Slope, Exercise Angina, Chest Pain Type, FastingBS, and the engineered features) provide a robust foundation for identifying individuals at higher risk of heart disease. This aligns with established cardiovascular risk factors reported at population level (American Heart Association Council on Epidemiology and Prevention Statistics Committee and Stroke Statistics Committee, 2026), and allows for proactive screening and intervention strategies.
2. **Targeted Patient Stratification:** The nuanced insights from features like Age_Group, BP_Category, Heart_Rate_Age_Ratio, and especially ST_Exercise_Risk enable more precise patient stratification. This means healthcare resources can be optimally allocated to those who need them most, rather than a one-size-fits-all approach.
3. **Improved Diagnostic Efficiency:** Understanding the strong associations between certain clinical indicators and heart disease can streamline diagnostic processes. For instance, a patient presenting with an 'Asymptomatic' chest pain type combined with a 'Flat_Yes' ST_Exercise_Risk should trigger immediate and thorough cardiovascular investigation.
4. **Personalized Patient Management:** The ability to quantify the impact of various factors on heart disease risk supports the development of personalised prevention and treatment plans. Clinicians can better explain risk to patients, fostering adherence to lifestyle modifications or medication.
5. **Validation of Clinical Intuition:** The statistical validation confirms the importance of many established clinical indicators, providing data-driven evidence to support existing medical guidelines while also highlighting the value of novel, data-driven feature combinations.

5. Recommendations

Based on the analysis, the following evidence-based recommendations are proposed:

1. **Prioritize Key Predictors in Clinical Assessment:** Integrate Age, MaxHR, Oldpeak, ST-Slope patterns, presence of Exercise Angina, and Chest Pain Type as primary considerations in initial patient assessments for heart disease risk. These should form the core of any intake questionnaire or preliminary diagnostic evaluation.
2. **Adopt Engineered Features for Comprehensive Risk Scoring:** Develop or integrate a risk scoring system that incorporates the engineered features, particularly Heart_Rate_Age_Ratio and ST_Exercise_Risk. These features offer strong predictive signal by capturing complex interactions and should be part of a more advanced risk calculator.
3. **Develop Targeted Screening Programs:** Design specific screening protocols for patient subgroups identified as high-risk by Age_Group (e.g., more frequent screenings for the Elderly and Middle-Aged) and BP_Category (e.g., intensified monitoring for individuals with Hypertension or Elevated BP).
4. **Educate Patients on Lifestyle Modifications:** Leverage the findings on FastingBS, RestingBP, and Cholesterol to emphasise the importance of lifestyle interventions (diet, exercise) for managing these factors, even where their individual correlation with heart disease was weaker in this specific dataset.
5. **Utilize Findings for Physician Training:** Incorporate these analytical insights into continuous medical education programmes to ensure clinicians are aware of the most statistically impactful predictors and how they combine to influence heart disease risk.
6. **Implement a Predictive Modelling System:** Proceed with the development and evaluation of a machine learning model using the refined Final_Modelling_Dataset. Candidate approaches include logistic regression and tree-based ensemble methods such as gradient boosting (Chen & Guestrin, 2016), selected and compared using the standard model-evaluation practices set out for supervised classification problems (James et al., 2021).

6. Ethical and Responsible AI Considerations

This section is added to the original analysis in view of the Week 4 transition towards predictive modelling. Because any model built on this dataset would ultimately inform decisions about individual patients' care, its development should be guided by a recognised risk-management framework rather than treated purely as a technical exercise.

The NIST AI Risk Management Framework provides practical guidance for managing risks associated with AI systems and promoting trustworthy and responsible AI development and use (Tabassi, 2023). Applied to this project, this means that before any model based on the Final_Modelling_Dataset is used to inform real clinical decisions, it should be validated on data beyond the current 918-patient sample, checked for performance differences across demographic subgroups such as Sex, and reviewed by qualified clinical professionals. The model should be positioned as a decision-support tool that assists, rather than replaces, clinical judgement.

7. Dataset Limitations

While the analysis provides valuable insights, it is important to acknowledge the following limitations of the dataset:

- **Sample Size:** The dataset contains 918 entries. While sufficient for initial analysis, a larger and more diverse dataset would enhance the generalisability and robustness of the findings, especially for detecting subtle patterns or rare conditions.
- **Data Source and Representativeness:** The origin of the dataset (e.g., single clinic, specific region) is not specified, which raises questions about its representativeness. Findings might not be fully generalisable to broader, diverse populations (different ethnicities, socioeconomic backgrounds, or geographical locations).
- **Potential Bias:** There may be inherent biases in data collection. For example, if the dataset primarily includes individuals who have already sought medical attention, it might underrepresent individuals with undiagnosed or asymptomatic heart disease.
- **Missing Information:** Although the current dataset is complete with no missing values after cleaning, the original raw data contained invalid zero values in RestingBP and Cholesterol that were imputed in Week 2. The methodology used to handle these values can influence the final data distribution and analytical outcomes.
- **Limited Feature Set:** While comprehensive for the features included, other potentially important factors influencing heart disease (e.g., family history, genetic predispositions, detailed dietary information, stress levels, socio-economic status, long-term medication use, other comorbidities) are not present in this dataset. The absence of these features might limit the predictive power and a holistic understanding of heart disease.
- **Cross-Sectional Nature:** The dataset appears to be cross-sectional, representing a snapshot in time. It does not provide longitudinal information, which would be crucial for understanding disease progression or the long-term impact of risk factors and interventions.
- **Categorisation Thresholds:** For engineered features like Age_Group and BP_Category, the thresholds used for categorisation are based on project-defined analytical bands informed by general medical guidance rather than a validated clinical classification. Different thresholds could yield slightly different insights or model performance.
- **Dependence on Clinical Measurements:** The dataset relies on clinical measurements and patient-reported symptoms. Subjectivity in symptom reporting (e.g., Chest Pain Type) or variability in measurement techniques could introduce noise.

8. Next Steps

The Final Modelling Dataset, now refined and thoroughly evaluated, is well-prepared for the next phase of the project: predictive modelling in Week 4. The following steps outline how this dataset will be utilised:

1. **Data Splitting:** The dataset will be divided into training, validation, and test sets. This ensures that the model can be trained on a portion of the data, fine-tuned on another, and evaluated on unseen data to provide an unbiased assessment of its performance.
2. **Model Selection:** Based on the nature of the data (a mix of numerical and categorical features and a binary target variable), appropriate machine learning models will be selected. Potential candidates include Logistic Regression, Support Vector Machines, Decision Trees, Random Forests, and Gradient Boosting Machines such as XGBoost (Chen & Guestrin, 2016).
3. **Model Training:** Selected models will be trained on the training dataset, learning the patterns and relationships between the features and the HeartDisease target variable.
4. **Hyperparameter Tuning:** Using the validation set, hyperparameters for each model will be optimised to achieve the best possible performance, preventing overfitting to the training data.
5. **Model Evaluation:** The performance of the optimised models will be rigorously evaluated on the unseen test set using metrics relevant to classification tasks, such as accuracy, precision, recall, F1-score, and ROC-AUC (James et al., 2021). This will assess the model's ability to generalise to new data.
6. **Feature Importance Analysis:** Post-modelling, an analysis of feature importance will be conducted to confirm which features contribute most significantly to the model's predictions. This can further validate the insights gained during the EDA and statistical testing phases.
7. **Interpretability and Explainability:** Efforts will be made to interpret the models, especially where complex models are used, to provide actionable insights into why certain predictions are made, consistent with the responsible-AI approach set out in Section 6 and important for clinical adoption (Tabassi, 2023).

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